

NLP

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Section: BDS-6A

Assignment

Task-1:

One-Hot Encoding

'The' → 1 0 0 0 0 0 0 0
'Students' → 0 1 0 0 0 0 0 0
'opened' → 0 0 1 0 0 0 0 0
'their' → 0 0 0 1 0 0 0 0
'laptops' → 0 0 0 0 1 0 0 0
'books' → 0 0 0 0 0 1 0 0
'exams' → 0 0 0 0 0 0 1 0
'minds' → 0 0 0 0 0 0 0 1

(1)

Embedding for Target Words

Embedding = $I \cdot W$

'The' → 3 2 3 4
'Students' → 0 3 1 4
'opened' → 3 2 0 2
'their' → 1 3 0 2
'laptops' → 3 2 0 0
'books' → 4 2 4 2
'exams' → 2 2 4 3
'minds' → 4 3 4 2

(2) Embedding for surrounding context words

'The' → 1 0 1 4
'students' → 1 1 1 1
'opened' → 2 0 2 2
'their' → 0 1 2 1
'laptops' → 1 4 0 1
'books' → 4 0 4 4
'exams' → 0 1 3 3
'minds' → 1 2 3 0

(3) Average Embedding

'The' → 2.1 2.4
'students' → 0.5 2.1 2.5
'opened' → 2.5 1.1 2
'their' → 0.5 2.1 1.5
'laptops' → 1.5 3 0.5
'books' → 4 1 4 3
'exams' → 1.1 1.5 3.5 3
'minds' → 2.5 2.5 3.5 1

Task-2:

The students opened their exam

LSTM with Target Word Embedding

$$h_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad c_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$I_1 = h_0 \cdot x_1$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 3 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$i_1 = \sigma(w_i \cdot I_1 + b_i)$$

$$= \sigma \left(\begin{bmatrix} 48 \\ 35 \\ 49 \\ 28 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 2 \end{bmatrix} \right)$$

$$= \sigma \begin{bmatrix} 48 \\ 37 \\ 50 \\ 30 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$f_1 = \sigma(w_f \cdot I_1 + b_f)$$

$$= \sigma \left(\begin{bmatrix} 17 \\ 47 \\ 21 \\ 26 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right)$$

$$= \sigma \begin{bmatrix} 17 \\ 47 \\ 22 \\ 32 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 1 \end{bmatrix}$$

$$\tilde{c}_1 = \tanh(w_c \cdot I_1 + b_c)$$

$$= \tanh \left(\begin{bmatrix} 35 \\ 39 \\ 35 \\ 51 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right) = \tanh \begin{bmatrix} 35 \\ 41 \\ 35 \\ 53 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$c_1 = f_1 \cdot c_0 + i_1 \cdot c_1$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 1 \end{bmatrix}$$

$$o_1 = \sigma (w_0 I_1 + b_0)$$

$$= \sigma \left(\begin{bmatrix} 31 \\ 26 \\ 31 \\ 54 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right) = \sigma \begin{bmatrix} 37 \\ 30 \\ 31 \\ 62 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$h_1 = o_1 \cdot c_1$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 1 \end{bmatrix}$$

$$I_2 = [h_1, x_2]$$

$$I_2 = \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 1 \\ 0 \\ 3 \\ 1 \\ 4 \end{bmatrix}$$

$$f_2 = \sigma(W_f I_2 + B_f)$$

$$= \sigma \left(\begin{bmatrix} 57.96 \\ 28.95 \\ 46.98 \\ 25.96 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right)$$

$$= \sigma \begin{bmatrix} 57.96 \\ 30.95 \\ 47.98 \\ 25.96 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$i_2 = \sigma(W_i I_2 + B_i)$$

$$= \sigma \left(\begin{bmatrix} 34.89 \\ 44.95 \\ 25.96 \\ 20.95 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right)$$

$$= \sigma \begin{bmatrix} 34.89 \\ 44.95 \\ 26.96 \\ 26.95 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\tilde{c}_2 = \tanh(W_c I_2 + B_c)$$

$$= \tanh \left(\begin{bmatrix} 34.94 \\ 47.95 \\ 43.95 \\ 41.92 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right)$$

$$= \tanh \begin{bmatrix} 34.94 \\ 49.95 \\ 43.95 \\ 43.92 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$C_2 = F_2 \cdot C_1 + i_2 \cdot C_2$$

$$= \begin{bmatrix} 1 & 0.99 \\ 1 & 1 \\ 1 & 0.99 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.99 \\ 1 & 1 \\ 1 & 0.99 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1.99 \\ 2 \\ 1.99 \\ 2 \end{bmatrix}$$

$$O_2 = \Sigma (W_0 I_2 + B_0)$$

$$= \Sigma \left(\begin{bmatrix} 43.92 \\ 17.94 \\ 30.93 \\ 48.91 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right) = \Sigma \begin{bmatrix} 49.92 \\ 21.94 \\ 30.93 \\ 56.91 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.99 \\ 1 \\ 1 \end{bmatrix}$$

$$h_2 = O_2 \cdot C_2$$

$$= \begin{bmatrix} 1 & 1.99 \\ 0.99 & 2 \\ 1 & 1.99 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1.99 \\ 1.98 \\ 1.99 \\ 2 \end{bmatrix}$$

$$I_3 = [h_2, x_3]$$

$$I_3 = \begin{bmatrix} 1.99 & 0 \\ 1.98 & 2 \\ 1.99 & \\ 2 & \\ 3 & \\ 2 & \end{bmatrix}$$

$$f_3 = \sigma (W_f \cdot I_3 + b_f)$$

$$= \sigma \left(\begin{array}{c|c} 47.92 & 0 \\ 27.93 & + 2 \\ 47.94 & 1 \\ 28.92 & 0 \end{array} \right) = \sigma \left(\begin{array}{c|c} 47.92 & 1 \\ 29.93 & 1 \\ 48.94 & 1 \\ 28.92 & 1 \end{array} \right)$$

$$i_3 = \sigma (W_i I_3 + B_i)$$

$$= \sigma \left(\begin{array}{c|c} 46.79 & 0 \\ 45.91 & + 0 \\ 25.92 & 1 \\ 41.87 & 6 \end{array} \right) = \sigma \left(\begin{array}{c|c} 46.79 & 1 \\ 45.91 & 1 \\ 26.92 & 1 \\ 47.87 & 1 \end{array} \right)$$

$$\tilde{C}_3 = \tanh (W_c I_3 + B_c)$$

$$= \tanh \left(\begin{array}{c|c} 50.84 & 0 \\ 41.93 & + 2 \\ 39.95 & 0 \\ 54.92 & 2 \end{array} \right) = \tanh \left(\begin{array}{c|c} 50.84 & 1 \\ 43.93 & 1 \\ 39.95 & 1 \\ 56.92 & 1 \end{array} \right)$$

$$C_3 = f_3 \cdot C_2 + i_3 \cdot \tilde{C}_3$$

$$= \left(\begin{array}{c|c} 1 & 1.99 \\ 1 & 2 \\ 1 & 1.99 \\ 1 & 2 \end{array} \right) + \left(\begin{array}{c|c} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{array} \right) = \left(\begin{array}{c|c} 1.99 & 1 \\ 2 & 1 \\ 1.99 & 1 \\ 2 & 1 \end{array} \right) = \left(\begin{array}{c|c} 2.99 & 3 \\ 2.99 & 3 \end{array} \right)$$

$$O_3 = \sigma (W_o I_3 + B_o)$$

$$= \sigma \left(\begin{array}{c|c} 50.84 & 6 \\ 36.9 & + 4 \\ 31.93 & 0 \\ 58.89 & 8 \end{array} \right) = \sigma \left(\begin{array}{c|c} 56.84 & 1 \\ 40.9 & 1 \\ 31.93 & 1 \\ 66.89 & 1 \end{array} \right)$$

$$h_3 = 03.03$$

$$= \begin{bmatrix} 1 & 2.99 \\ 1 & 3 \\ 1 & 2.99 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 2.99 \\ 3 \\ 2.99 \\ 3 \end{bmatrix}$$

$$I_4 = [h_3 \cdot \lambda_4]$$

$$= \begin{bmatrix} 2.99 \\ 3 \\ 2.99 \\ 3 \\ 1 \\ 3 \\ 0 \\ 2 \end{bmatrix}$$

$$F_4 = \sigma(W_f I_4 + B_f)$$

$$= \sigma \left(\begin{bmatrix} 65.96 \\ 29.95 \\ 51.98 \\ 37.96 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right) = \sigma \begin{bmatrix} 65.96 \\ 31.95 \\ 52.98 \\ 37.96 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$i_4 = \sigma(W_i I_4 + B_i)$$

$$= \sigma \left(\begin{bmatrix} 65.89 \\ 55.95 \\ 36.96 \\ 48.95 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right) = \sigma \begin{bmatrix} 65.89 \\ 55.95 \\ 37.96 \\ 54.95 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\tilde{C}_4 = \tanh(W_c I_4 + B_c)$$

$$= \tanh \left(\begin{array}{c|c} 55.94 & 0 \\ 56.95 & 2 \\ 52.95 & 0 \\ 62.92 & 2 \end{array} \right) = \tanh \left(\begin{array}{c|c} 55.94 & 1 \\ 58.95 & 1 \\ 52.95 & 1 \\ 64.92 & 1 \end{array} \right)$$

$$C_4 = f_4 \cdot C_3 + i_4 \cdot \tilde{C}_4$$

$$= \begin{array}{c|c} 1 & 2.99 \\ 1 & 3 \\ 1 & 2.99 \\ 1 & 3 \end{array} + \begin{array}{c|c} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{array} = \begin{array}{c|c} 2.99 & 1 \\ 3 & 1 \\ 2.99 & 1 \\ 3 & 1 \end{array} = \begin{array}{c|c} 3.99 & 1 \\ 4 & 1 \\ 3.99 & 1 \\ 4 & 1 \end{array}$$

$$O_4 = \sigma(W_o I_4 + B_o)$$

$$= \sigma \left(\begin{array}{c|c} 68.92 & 6 \\ 35.94 & 4 \\ 35.93 & 0 \\ 64.91 & 8 \end{array} \right) = \sigma \left(\begin{array}{c|c} 74.92 & 1 \\ 39.94 & 1 \\ 35.93 & 1 \\ 68.91 & 1 \end{array} \right)$$

$$h_4 = O_4 \cdot C_4$$

$$= \begin{array}{c|c} 1 & 3.99 \\ 1 & 4 \\ 1 & 3.99 \\ 1 & 4 \end{array} = \begin{array}{c|c} 3.99 & 1 \\ 4 & 1 \\ 3.99 & 1 \\ 4 & 1 \end{array}$$

$$\hat{y} = \text{softmax}(h_4^{(T)} W + B)$$

$$\hat{h}_4 = [3.99 \ 4 \ 3.99 \ 4]$$

$$\hat{y} = \text{softmax}$$

55.88	3	softmax	58.88
35.95	2		37.95
59.91	0		59.91
39.92	0		39.92
55.94	3		58.94
43.96	0		43.96
63.94	3		66.94
55.95	0		55.95

$$\hat{y} =$$

0.0003
2.5×10^{-13}
0.0004
1.8×10^{-12}
0.0003
1×10^{-10}
0.99
1.6×10^{-5}

→ max → exams

↓
predicted word

The students opened their exams

LSTM with context word embedding

$$h_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$c_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$I_1 = [h_0, x_1]$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 1 \\ 4 \end{bmatrix}$$

$$f_1 = \sigma(W_f I_1 + B_f)$$

$$= \sigma \left(\begin{bmatrix} 28 \\ 25 \\ 29 \\ 14 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 28 \\ 27 \\ 30 \\ 14 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0.99 \end{bmatrix}$$

$$i_1 = \sigma(W_i I_1 + B_i)$$

$$= \sigma \left(\begin{bmatrix} 5 \\ 31 \\ 11 \\ 4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 5 \\ 31 \\ 12 \\ 10 \end{bmatrix} \right) = \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 0.99 \end{bmatrix}$$

$$\tilde{c}_1 = \tanh(W_c I_1 + B_c)$$

$$= \tanh \left(\begin{bmatrix} 17 \\ 27 \\ 19 \\ 19 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right) = \tanh \begin{bmatrix} 17 \\ 29 \\ 19 \\ 21 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$c_1 = f_1 \cdot C_0 + i_1 \cdot C_1$$

$$= \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0.99 & 0 \end{bmatrix} + \begin{bmatrix} 0.99 & 1 \\ 1 & 1 \\ 0.99 & 1 \\ 0.99 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0.99 \\ 0 & 1 \\ 0 & 0.99 \\ 0 & 0.99 \end{bmatrix}$$

$$= \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 0.99 \end{bmatrix}$$

$$o_1 = \sigma(W_0 I + B_0)$$

$$= \sigma \left(\begin{bmatrix} 19 \\ 14 \\ 23 \\ 32 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right) = \sigma \begin{bmatrix} 25 \\ 18 \\ 23 \\ 40 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.99 \\ 0.99 \\ 1 \end{bmatrix}$$

$$h_1 = o_1 \cdot c_1$$

$$= \begin{bmatrix} 1 \\ 0.99 \\ 0.99 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 0.99 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 0.99 \\ 0.98 \\ 0.99 \end{bmatrix}$$

$$I_2 = [x_1, x_2]$$

$$I_2 = \begin{bmatrix} 0.99 \\ 0.99 \\ 0.98 \\ 0.99 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$f_2 = \sigma (W_2 I_2 + B_f)$$

$$= \sigma \left(\begin{bmatrix} 27.92 \\ 15.93 \\ 22.95 \\ 18.92 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 27.92 \\ 17.93 \\ 23.95 \\ 18.92 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0.99 \\ 1 \\ 0.99 \end{bmatrix}$$

$$i_2 = \sigma (W_i I_2 + B_i)$$

$$= \sigma \left(\begin{bmatrix} 27.79 \\ 26.91 \\ 15.93 \\ 22.9 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 27.79 \\ 26.91 \\ 16.93 \\ 28.9 \end{bmatrix} \right) = \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 1 \end{bmatrix}$$

$$\tilde{c}_2 = \tanh (W_c I_2 + B_c)$$

$$= \tanh \left(\begin{bmatrix} 22.83 \\ 23.89 \\ 21.9 \\ 30.86 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right) = \tanh \left(\begin{bmatrix} 22.83 \\ 25.89 \\ 21.9 \\ 32.86 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$c_2 = f_2 \cdot c_1 + i_2 \cdot \tilde{c}_2$$

$$= \begin{bmatrix} 1 \\ 0.99 \\ 1 \\ 0.99 \end{bmatrix} \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 0.99 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.98 \\ 1.99 \\ 1.98 \\ 1.99 \end{bmatrix}$$

$$O_2 = \sigma(W_0 I_2 + B_0)$$

$$= \sigma \left(\begin{bmatrix} 26.85 \\ 16.9 \\ 15.89 \\ 29.85 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 32.85 \\ 20.9 \\ 15.89 \\ 37.85 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0.99 \\ 0.99 \\ 1 \end{bmatrix}$$

$$h_2 = O_2 \cdot C_2$$

$$= \begin{bmatrix} 1 \\ 0.99 \\ 0.99 \\ 1 \end{bmatrix} \begin{bmatrix} 1.98 \\ 1.99 \\ 1.98 \\ 1.99 \end{bmatrix} = \begin{bmatrix} 1.98 \\ 1.97 \\ 1.96 \\ 1.99 \end{bmatrix}$$

$$I_3 = [h_2, x_3]$$

$$= \begin{bmatrix} 1.98 \\ 1.97 \\ 1.96 \\ 1.99 \\ 2 \\ 0 \\ 2 \\ 2 \end{bmatrix}$$

$$f_3 = \sigma(W_f I_3 + B_f)$$

$$= \sigma \left(\begin{bmatrix} 43.76 \\ 31.85 \\ 35.85 \\ 35.78 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 43.76 \\ 33.85 \\ 36.85 \\ 35.78 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$i_3 = \sigma (W_i I_3 + B_i)$$

$$= \sigma \left(\begin{bmatrix} 35.53 \\ 51.74 \\ 27.81 \\ 43.71 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 35.53 \\ 51.74 \\ 28.81 \\ 49.71 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\tilde{C}_3 = \tanh (W_c I_3 + B_c)$$

$$= \tanh \left(\begin{bmatrix} 37.61 \\ 41.71 \\ 33.75 \\ 51.67 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right) = \tanh \left(\begin{bmatrix} 37.61 \\ 43.71 \\ 33.75 \\ 53.67 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$C_3 = f_3 \cdot C_2 + i_3 \cdot \tilde{C}_3$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1.98 \\ 1.99 \\ 1.98 \\ 1.99 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.98 \\ 1.99 \\ 1.98 \\ 1.99 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.98 \\ 2.99 \\ 2.98 \\ 2.99 \end{bmatrix}$$

$$O_3 = \sigma (W_o I_3 + B_o)$$

$$= \sigma \left(\begin{bmatrix} 47.61 \\ 33.77 \\ 29.78 \\ 53.66 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 53.61 \\ 37.77 \\ 29.78 \\ 61.66 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$h_3 = 0.3 \cdot C_3$$

$$= \begin{bmatrix} 1 & 2.98 \\ 1 & 2.99 \\ 1 & 2.98 \\ 1 & 2.99 \end{bmatrix} = \begin{bmatrix} 2.98 \\ 2.99 \\ 2.98 \\ 2.99 \end{bmatrix}$$

$$I_4 = [h_3, \lambda_4]$$

$$I_4 = \begin{bmatrix} 2.98 \\ 2.99 \\ 2.98 \\ 2.99 \\ 0 \\ 1 \\ 2 \\ 1 \end{bmatrix}$$

$$f_4 = \sigma(W_f I_4 + B_f)$$

$$= \sigma \left(\begin{bmatrix} 55.84 \\ 28.89 \\ 33.91 \\ 42.86 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 55.84 \\ 30.89 \\ 34.91 \\ 42.86 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$i_4 = \sigma(W_i I_4 + B_i)$$

$$= \sigma \left(\begin{bmatrix} 53.73 \\ 55.82 \\ 36.87 \\ 50.81 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 53.73 \\ 55.82 \\ 37.87 \\ 56.81 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

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$$C_4 = \tanh(W_c I_4 + B_c)$$

$$\tilde{C}_4 = \tanh \left(\begin{bmatrix} 39.83 \\ 50.83 \\ 44.85 \\ 57.79 \end{bmatrix} + \begin{bmatrix} -0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right) = \tanh \left(\begin{bmatrix} 39.83 \\ 52.83 \\ 44.85 \\ 59.79 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$C_4 = f_4 \cdot C_3 + i_4 \cdot \tilde{C}_4$$

$$= \begin{bmatrix} 1 & 2.98 & 1 & 1 \\ 1 & 2.99 & 1 & 1 \\ 1 & 2.98 & 1 & 1 \\ 1 & 2.99 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2.98 & 1 & 3.98 \\ 2.99 & 1 & 3.99 \\ 2.98 & 1 & 3.98 \\ 2.99 & 1 & 3.99 \end{bmatrix}$$

$$O_4 = \sigma(W_{O4} + B_O)$$

$$= \sigma \left(\begin{bmatrix} 61.75 \\ 30.85 \\ 28.86 \\ 53.78 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 67.75 \\ 34.85 \\ 28.86 \\ 61.78 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$H_4 = O_4 \cdot C_4$$

$$= \begin{bmatrix} 1 & 3.98 \\ 1 & 3.99 \\ 1 & 3.98 \\ 1 & 3.99 \end{bmatrix} = \begin{bmatrix} 3.98 \\ 3.99 \\ 3.98 \\ 3.99 \end{bmatrix}$$

$$\hat{y} = \text{softmax}(W_{hy} + B)$$

$$= \text{softmax} \left(\begin{bmatrix} 55.74 \\ 35.86 \\ 59.76 \\ 39.82 \\ 55.8 \\ 43.85 \end{bmatrix} + \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \\ 3 \\ 0 \end{bmatrix} \right)$$

$$\begin{bmatrix} 63.78 \\ 55.81 \end{bmatrix} + \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$g = \text{soft max}$

58.74	0.0003	predicted word ↓ max → exams
37.86	2.7×10^{-13}	
59.76	0.00089	
39.82	1.9×10^{-12}	
58.8	0.0003	
43.85	1.09×10^{-10}	
66.78	0.99	
55.81	1.7×10^{-5}	

The students opened their exams

LSTM with average embedding

$$h_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$c_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$I_1 = [h_0, x_1]$$

$$I_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 2 \\ 1 \\ 2 \\ 4 \end{bmatrix}$$

$$f_1 = \sigma(W_f I_1 + B_f)$$

$$= \sigma \left(\begin{bmatrix} 38 \\ 30 \\ 29 \\ 21 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 38 \\ 32 \\ 30 \\ 21 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0.99 \end{bmatrix}$$

$$i_1 = \sigma(W_i I_1 + B_i)$$

$$= \sigma \left(\begin{bmatrix} 11 \\ 39 \\ 16 \\ 17 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 11 \\ 39 \\ 17 \\ 23 \end{bmatrix} \right) = \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 0.99 \end{bmatrix}$$

$$\tilde{C}_1 = \tanh(W_c I_1 + B_c)$$

$$= \tanh \left(\begin{bmatrix} 26 \\ 33 \\ 27 \\ 25 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right) = \tanh \left(\begin{bmatrix} 26 \\ 35 \\ 27 \\ 27 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$C_1 = f_1 \cdot C_0 + i_1 \cdot \tilde{C}_1$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0.99 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 0.99 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 0.99 \end{bmatrix}$$

$$= \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 0.99 \end{bmatrix}$$

$$O_1 = \sigma(W_0 I_1 + b_0)$$

$$= \sigma \left(\begin{bmatrix} -25 \\ 20 \\ 27 \\ 43 \end{bmatrix} + \begin{bmatrix} -6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right) = \sigma \begin{bmatrix} -31 \\ 24 \\ 27 \\ 51 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$h_1 = O_1 \cdot c_1$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 0.99 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 0.99 \end{bmatrix}$$

$$I_2 = [h_1, x_2]$$

$$I_2 = \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 0.99 \\ 0.5 \\ 2 \\ 0.1 \\ 2.5 \end{bmatrix}$$

$$f_2 = \sigma(W_f \cdot I_2 + B_f)$$

$$= \sigma \left(\begin{bmatrix} 42.9 \\ 22.45 \\ 34.95 \\ 22.42 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right) = \sigma \begin{bmatrix} 42.9 \\ 24.45 \\ 35.95 \\ 22.42 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0.99 \end{bmatrix}$$

$$i_2 = \sigma (W_i I_2 + B_i)$$

$$= \sigma \left(\begin{bmatrix} 28.89 \\ 35.89 \\ 20.93 \\ 21.9 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 28.89 \\ 35.89 \\ 20.93 \\ 27.9 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 0.99 \\ 1 \end{bmatrix}$$

$$\bar{C}_2 = \tanh (W_c I_2 + B_c)$$

$$= \tanh \left(\begin{bmatrix} 28.94 \\ 35.89 \\ 32.9 \\ 36.37 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right) = \tanh \left(\begin{bmatrix} 28.94 \\ 37.89 \\ 32.9 \\ 38.37 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$C_2 = f_2 \cdot C_1 + i_2 \cdot \bar{C}_2$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0.99 \end{bmatrix} \cdot \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 0.99 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0.99 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0.99 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 1 \\ 0.99 \\ 0.98 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0.99 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1.99 \\ 2 \\ 1.98 \\ 1.98 \end{bmatrix}$$

$$o_2 = \sigma (W_o I_2 + B_o)$$

$$= \sigma \left(\begin{bmatrix} 35.37 \\ 17.43 \\ 23.43 \\ 39.38 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 41.37 \\ 21.43 \\ 23.43 \\ 47.38 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0.99 \\ 0.99 \\ 1 \end{bmatrix}$$

$$h_2 = 02.02$$

$$= \begin{bmatrix} 1 & 1.99 \\ 0.99 & 2 \\ 0.99 & 1.98 \\ 1 & 1.98 \end{bmatrix} = \begin{bmatrix} 1.99 \\ 1.98 \\ 1.96 \\ 1.98 \end{bmatrix}$$

$$I_3 = [h_2, x_3]$$

$$I_3 = \begin{bmatrix} 1.99 \\ 1.98 \\ 1.96 \\ 1.98 \\ 2.5 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

$$f_3 = \sum (w_i I_3 + B_i)$$

$$= \sum \begin{bmatrix} 45.74 \\ 29.9 \\ 41.85 \\ 32.28 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} = \sum \begin{bmatrix} 45.74 \\ 31.9 \\ 42.85 \\ 32.28 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$i_3 = \sum (w_i I_3 + B_i)$$

$$= \sum \begin{bmatrix} 41.14 \\ 48.73 \\ 26.83 \\ 42.74 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} = \sum \begin{bmatrix} 41.14 \\ 48.73 \\ 27.83 \\ 48.74 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\tilde{C}_3 = \tanh(WcI_3 + Bc)$$

$$= \tanh \left(\begin{bmatrix} 44.16 \\ 41.66 \\ 35.7 \\ 53.14 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right) = \tanh \begin{bmatrix} 44.16 \\ 43.66 \\ 35.7 \\ 55.14 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$C_3 = f_3 \cdot C_2 + i_3 \cdot \tilde{C}_3$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1.99 \\ 2 \\ 1.98 \\ 1.98 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1.99 \\ 2 \\ 1.98 \\ 1.98 \end{bmatrix}$$

$$= \begin{bmatrix} 2.99 \\ 3 \\ 2.98 \\ 2.98 \end{bmatrix}$$

$$O_3 = \sigma(WoI_3 + B_o)$$

$$= \sigma \left(\begin{bmatrix} 49.15 \\ 35.32 \\ 30.81 \\ 56.18 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right) = \sigma \begin{bmatrix} 55.15 \\ 39.32 \\ 30.81 \\ 64.18 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$h_3 = O_3 \cdot C_3$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 2.99 \\ 3 \\ 2.98 \\ 2.98 \end{bmatrix} = \begin{bmatrix} 2.99 \\ 3 \\ 2.98 \\ 2.98 \end{bmatrix}$$

$$I_4 = [h_3; x_4]$$

$$I_4 = \begin{bmatrix} 2.99 \\ 3 \\ 2.98 \\ 2.98 \\ 0.5 \\ 2 \\ 1 \\ 1.5 \end{bmatrix}$$

$$f_4 = \sigma (W_f I_4 + B_f)$$

$$= \sigma \left(\begin{bmatrix} 60.82 \\ 29.44 \\ 42.91 \\ 40.36 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 60.82 \\ 31.44 \\ 43.91 \\ 40.36 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$i_4 = \sigma (W_i I_4 + B_i)$$

$$= \sigma \left(\begin{bmatrix} 59.84 \\ 55.81 \\ 36.89 \\ 49.84 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right) = \sigma \left(\begin{bmatrix} 59.84 \\ 55.81 \\ 37.89 \\ 55.84 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\tilde{C}_4 = \tanh (W_c I_4 + B_c)$$

$$= \tanh \left(\begin{bmatrix} 47.88 \\ 53.78 \\ 48.8 \\ 60.26 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right) = \tanh \left(\begin{bmatrix} 47.88 \\ 55.78 \\ 48.8 \\ 62.26 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$C_4 = f_4 \cdot C_3 + i_4 \cdot \tilde{C}_3$$

$$= \begin{bmatrix} 1 & 2.99 \\ 1 & 3 \\ 1 & 2.98 \\ 1 & 2.98 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2.99 & 1 \\ 3 & 1 \\ 2.98 & 1 \\ 2.98 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3.99 \\ 4 \\ 3.98 \\ 3.98 \end{bmatrix}$$

$$O_4 = \sigma(W O_4 + B_0)$$

$$= \sigma \left(\begin{bmatrix} 65.29 \\ 33.4 \\ 32.39 \\ 59.3 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right) = \sigma \begin{bmatrix} 71.29 \\ 37.4 \\ 32.39 \\ 67.3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$h_4 = O_4 \cdot C_4$$

$$= \begin{bmatrix} 1 & 3.99 \\ 1 & 4 \\ 1 & 3.98 \\ 1 & 3.98 \end{bmatrix} = \begin{bmatrix} 3.99 \\ 4 \\ 3.98 \\ 3.98 \end{bmatrix}$$

$$\hat{y} = \text{softmax}(W h_4 + B)$$

= softmax	55.82	3	= softmax	58.82
	35.89	2		37.89
	59.85	0		59.85
	39.88	0		39.88
	55.86	3		58.86
	43.84	0		43.84
	63.8	3		66.8
	55.79	2		57.79

=	0.0003
	2.7×10^{-13}
	0.0009
	2×10^{-12}
	0.0003
	1×10^{-10}
	0.99 → max → exams
	1.2×10^{-4}

↓
predicted word

The students opened their exams